OPEN AGENT HACKATHON 2026 · TRACK 2: THE AGENT THAT CAN EXPLAIN WHY

RecallGuard

Agents that check every item on a pharmacy shelf against FDA drug recalls, prove which lots are affected, and remember what was already handled.

Zetaris data layer

Meterless H-MEM

Nightshift + NVIDIA Nemotron

Real openFDA data

THE PROBLEM

Matching FDA recalls to the shelf is manual, slow, and easy to get wrong

1,436

drug recall reports since Jan 2025 in openFDA, each listing NDCs and lots in free text

3 formats

FDA codes are 10-digit (4-4-2, 5-3-2, 5-4-1); pharmacy systems store 11 digits with no dashes

0 memory

nothing tracks which recalls staff already pulled, so every check starts from scratch

WHO IT’S FOR

The pharmacist at an independent community pharmacy

No recall team

Chains have compliance staff and integrated systems. A small pharmacy has one pharmacist, a shelf export and the FDA website.

High stakes either way

Missing a Class I lot puts patients at risk. Over-pulling on a name match wastes stock and time.

Needs proof, not a guess

Every action must point to the FDA record and the exact lot text that justifies it.

Weekly, not once

Recalls keep arriving. Each check should surface only what is new or still open.

THE SOLUTION

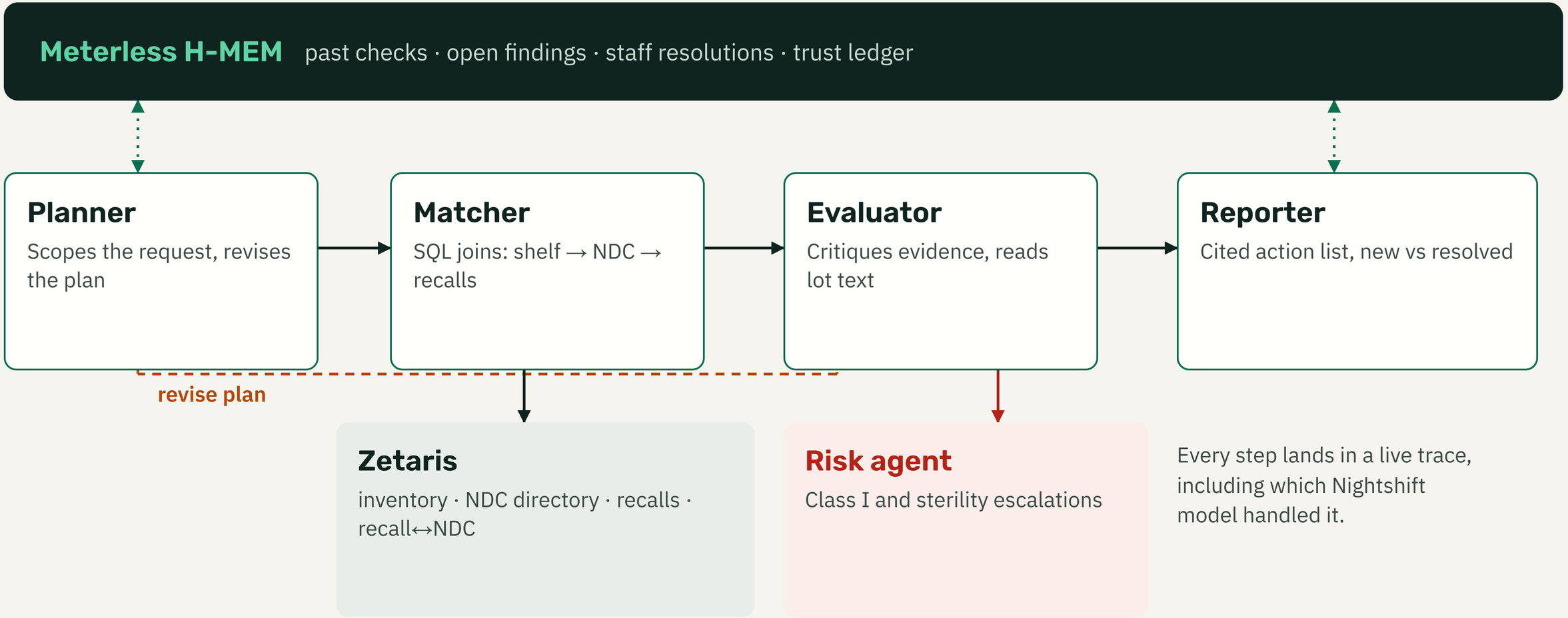
One request in, a cited action list out

Action	Shelf item · lot	Recall	Why
PULL NOW	Homeopathic nasal spray · X240137	D-0289-2026 · Class I	Recall covers all lots
PULL NOW	Bisoprolol/HCTZ · GBHL24005A	D-0301-2026	Lot named; nitrosamine impurity
QUARANTINE	Pegfilgrastim · 2199821	D-0353-2026 · Class I	Lot named, only some serials recalled
CLEARED	Clevipidine · X240211	D-0689-2026	Product recalled, our lot is not listed

Real recalls from openFDA; synthetic shelf inventory. Each row links to the FDA record and quotes the lot text.

AGENT ARCHITECTURE

Five agents in a loop, not a pipeline



Every step lands in a live trace, including which Nightshift model handled it.

THE FEEDBACK LOOP, FROM A REAL TRACE

The bisoprolol recall is only found because the Evaluator pushed back

matcher	18 of 19 shelf NDCs resolved; **SKU-1003 missing** from the FDA NDC directory
evaluator	“Missing evidence: 1 item could not be resolved. Requesting plan revision.”
planner	Adds **infer_product_ndc** and **match_by_name** steps; starts iteration 2
matcher	Infers product 29300-187 from the 11-digit code; links recall D-0301-2026
evaluator	Lot GBHL24005A named in recall → **PULL NOW**, confidence 0.80; escalated for nitrosamine impurity

EXPLAINING WHY

The Evaluator reads the FDA’s free-text lot field and quotes it

> “All lots within expiry.”

All lots → pull now

Our lot X240137 isn’t named, but the recall covers every lot.

> “Subset of Lot 2199821 … Specific Serial Numbers being Recalled”

Partial → quarantine

Hold the units and check each serial number before releasing any.

> “Lot #: 1239521, 1237454, Exp. Date 10/2027”

Not listed → cleared

Same product, but our lot X240211 isn’t recalled. No stock wasted.

On Class I or ambiguous text, a Nemotron model cross-checks the rules. If they disagree, the item is downgraded to manual verification, never silently decided.

MEMORY ACROSS SESSIONS · METERLESS H-MEM

Next week’s check only raises what is new or still open

Session 1

5 findings, all **NEW**. Each is mined into H-MEM with the FDA recall as its source.

Pharmacist marks bisoprolol resolved: “returned to supplier, credit memo CM-4471”.

`stored as user_correction · long_term · derivedFrom the finding`

→

Session 2

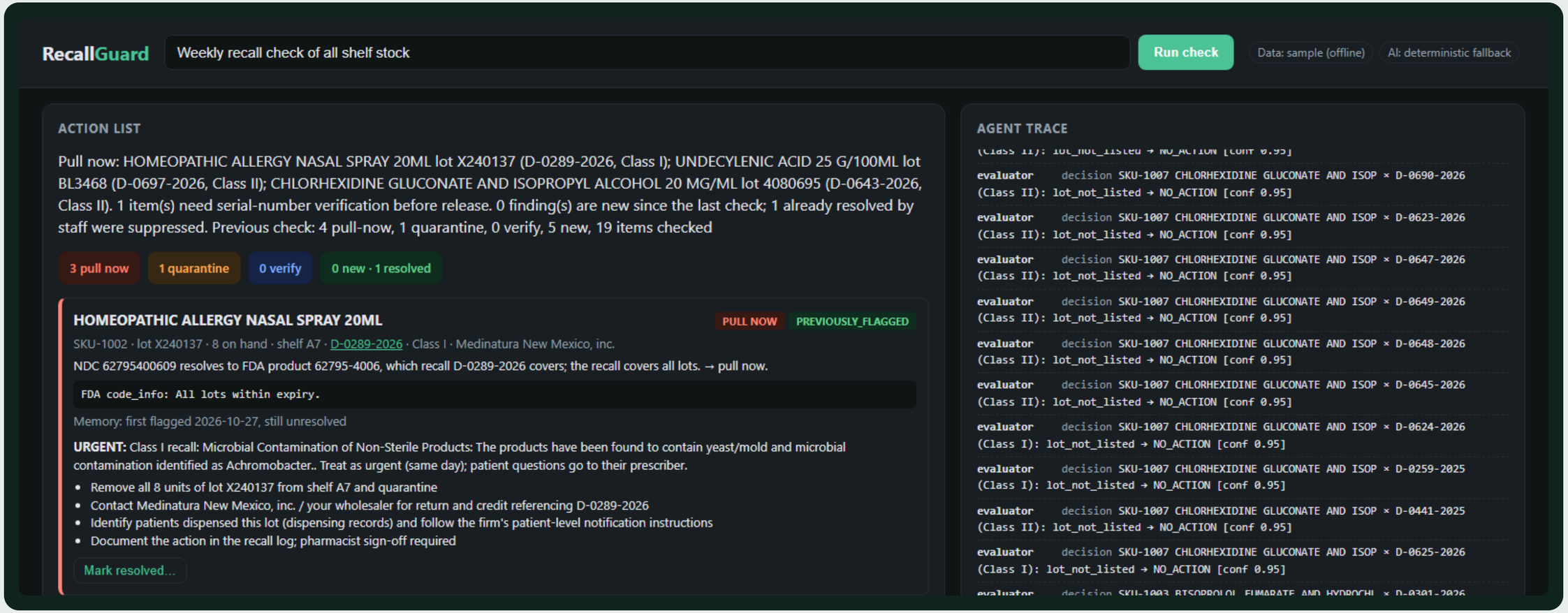
Bisoprolol: **ALREADY RESOLVED**, alert suppressed, note shown.

4 others: **PREVIOUSLY FLAGGED**, still unresolved since 25 Oct.

Briefing gets H-MEM’s ranked reinjection context and the last check’s summary.

THE PRODUCT

Action cards on the left, the agents' live trace on the right



SPONSOR TECHNOLOGY

Each sponsor carries a load-bearing part of the system

Technology	Role in RecallGuard	Proof it works
Zetaris	Data layer: all agent SQL goes through its REST query API, joining 4 sources	ZetarisLayer with login, paging, close; health check plus sample fallback
Meterless H-MEM	Memory of findings and staff resolutions across sessions; reinjection into the briefing	Official conformance suite 13/13; cross-session test
Meterless Nightshift	AI provider, OpenAI-compatible, Runtime 0.90.0 contract	Model per call logged in the trace
NVIDIA Nemotron	Lot-scope cross-check, request scoping, risk note, briefing	Routed through Nightshift Providers

WHAT'S DIFFERENT

It refuses to guess. Weak evidence goes back for more, and disagreement goes to a human.

Evidence ladder: exact package NDC → inferred product code → name + firm; name-only matches are rejected

Rules + model: deterministic lot parsing, with Nemotron as a second reader on high-stakes text

Memory as governance: staff corrections in H-MEM change what future runs report

TECH STACK AND RELIABILITY

Small, typed, and built to keep running when a service is down

Stack

- Node.js + TypeScript, zero framework
- Zetaris SQL REST API · local SQL engine for sample mode
- Meterless H-MEM engine · Nightshift client
- Web UI with live trace (server-sent events) + CLI

Reliability

- Zetaris down → cached openFDA sample, flagged in the trace
- No model → deterministic agents, same SQL
- Bounded loop (3 iterations), timeouts everywhere
- 6 app tests + 13/13 H-MEM conformance checks

RESULTS ON THE SAMPLE PHARMACY

Every affected item caught, every unaffected lot cleared

5 / 5

affected shelf items flagged with the right action

17

product-level recall matches correctly cleared by lot

1,436

real FDA recall reports checked against 19 shelf items

<1 s

full check in sample mode, with the complete trace

Measured on curated synthetic inventory against real openFDA data; expected outcomes are asserted in the test suite. Live-mode timing depends on Zetaris and the model.

FUTURE ENHANCEMENTS

From a weekly check to an always-on safety net

Live recall feed in Zetaris

Register the openFDA API as a Zetaris web table and alert the same day a recall is posted.

Dispensing records

Join recalled lots to dispensing history, so the patient-notification list is generated, not searched for.

Serial-level verification

Scan barcodes and check serials against the recall list, closing the quarantine loop automatically.

Shared memory across sites

Use the World Model engine so a group of pharmacies shares one recall state.

THANK YOU

RecallGuard

The right lot, pulled the same day, with the FDA evidence attached.

`github.com/axadishaq/recallguard`

Built solo by Asad for Open Agent Hackathon 2026 · Decision support only, not medical advice